

Thermodynamic State Vector Interface & Exergy Destruction Mechanics in Planetary Biosphere Simulation: Sprint 020 Report

Lead Scientific Communications & Academic Outreach Agent
Web of Life Research Initiative

Sprint 020 – Official Technical Report

Abstract

Planetary-scale ecological and biogeochemical simulations frequently suffer from physical inconsistencies, such as violations of energy conservation or entropy generation bounds over extended simulation horizons. In Sprint 020, the Web of Life simulation engine introduces a rigorous *Thermodynamic State Vector Interface* (`src/thermodynamics/types.ts`) and monad processing framework (`src/thermodynamics/thermodynamic_monad_process.ts`). This architecture mathematically couples elemental biogeochemical cycles (Carbon, Nitrogen, Phosphorus, Water) with strict First and Second Law thermodynamic constraints. By formalizing internal entropy generation ($\dot{S}_{\text{gen}} \geq 0$) and computing exergy destruction rates via the Gouy-Stodola theorem ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), the engine guarantees that all metabolic, radiative, and biochemical transformations maintain absolute thermodynamic validity.

1 Introduction & Thermodynamic Foundations

The Web of Life engine models planetary ecosystems as open thermodynamic pods driven exclusively by solar radiative influx ($\dot{Q}_{\text{solar}} > 0$). To prevent unphysical energy generation or entropy destruction within simulated metabolic networks, Sprint 020 establishes strict programmatic contracts enforcing classical thermodynamics.

1.1 First Law of Conservation (Energy)

For any subsystem V , the time rate of change of total internal energy U equals net energy fluxes across boundaries plus internal heat and work rates:

$$\frac{dE_{\text{sys}}}{dt} = \sum_j \dot{Q}_j - \dot{W} + \sum_i \dot{m}_i \left(h_i + \frac{1}{2} v_i^2 + g z_i \right) \quad (1)$$

1.2 Second Law of Thermodynamics (Entropy Generation)

The entropy balance for an open thermodynamic system is given by:

$$\frac{dS_{\text{sys}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \sum_i \dot{m}_i s_i + \dot{S}_{\text{gen}} \quad (2)$$

The core requirement of the Second Law is non-negative entropy generation:

$$\dot{S}_{\text{gen}} \geq 0 \quad (3)$$

where $\dot{S}_{\text{gen}} = 0$ represents reversible ideal processes, and $\dot{S}_{\text{gen}} > 0$ represents real irreversible processes (e.g., biochemical respiration, nutrient cycling).

1.3 Exergy Destruction Rate ($\dot{I}$)

Exergy (Ξ) represents the maximum useful work obtainable as the system brings itself into equilibrium with a reference environment at temperature $T_0 = 288.15$ K. The exergy destruction rate is directly proportional to internal entropy generation via the Gouy-Stodola theorem:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (4)$$

2 Interface Contracts & Monad Execution

The TypeScript architecture formalizes these equations into executable contracts (`src/thermodynamics/types` and `src/thermodynamics/thermodynamic_monad_process.ts`).

- **ThermodynamicBoundaryFlux:** Tracks boundary heat fluxes, temperatures, mass flow rates, and specific enthalpies/entropies.
- **ThermodynamicStateVector:** Encapsulates internal energy, entropy, reference temperature, entropy generation rate ($\dot{S}_{\text{gen}}$), exergy destruction rate ($\dot{I}$), and boundary fluxes.
- **Monad Transition Validation:** Evaluates $\dot{S}_{\text{gen}}$ dynamically upon every state transition, throwing a runtime violation error if $\dot{S}_{\text{gen}} < 0$.

3 Conclusion & Verification

Sprint 020 ensures that the Web of Life simulation engine operates within strict physical boundaries. Verification test suites confirm invariant adherence to the Gouy-Stodola relation and mass conservation across all biogeochemical monad stocks.

For complete source code, RFC specifications, and test suites, visit the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.